(51)

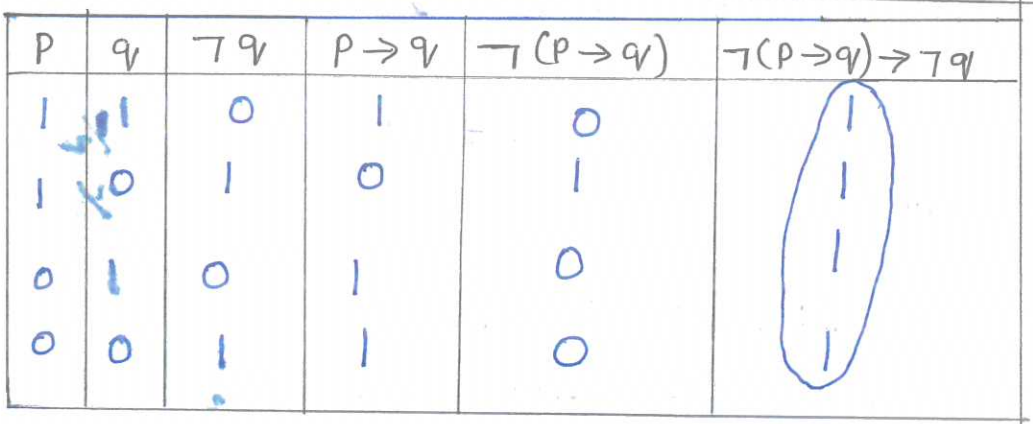
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Scoring Indicators

COURSE NAME :Discrete Mathematics

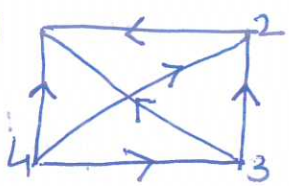
COURSE CODE :3341 **QID :** 2109230383

Q No	Scoring Indicators	Split score	Sub Total	Total score
	PART A			
I 1.	If you fail, then I fail	1	1	
I 2.	$P \leftrightarrow q \Leftrightarrow (P \to q) \wedge (q \to P)$ $\Leftrightarrow (\neg P \vee q) \wedge (\neg q \vee P)$ Dual of $P \leftrightarrow q$ is $(\neg P \wedge q) \vee (\neg q \wedge P)$	½. ½	1	
I 3.	$A = \{1, 2, 3\}$ (Give mark to any finite set)	1	1	
I 4.	$x^2 - 1 = 3 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$ $y + 3 = 4 \Rightarrow y = 4 - 3 \Rightarrow y = 1$	½ ½	1	9
I 5.	A function $f: A \to B$ is injective or one to one, if for every $b \in B$, there exist atleast one $a \in A$ such that $f(a) = b$.	1	1	
I 6	The Pigeon hole principle states that if n items are put into m containers with $n > m$, then atleast one container must contain more than one item.	1	1	
I 7	$20 + 35 + 15 = 70$ ways a Student can choose	1	1	

I 8.	A Semigroup is an algebraic structure consisting of a set together with an associative binary operation eg: Empty set.	1	1	
I 9.	A lattice L is a partially ordered set in which every pair of elements has a least upper bound (LUB) and a greatest lower bound (GLB)			
II	PART B			8×3
II 1.				

P	q	¬q	P→q	¬(P→q)	¬(P→q)→¬q
1	1	0	1	0	1
1	0	1	0	1	1
0	1	0	1	0	1
0	0	1	1	0	1

	The truth values of the last column are all 1's (truth). Hence it is a tautology.	2 1	3
II 2.	1. $P \wedge Q$ (premise) 2. Q (Conjunctive Simplification) 3. $\neg P \vee Q$ (disjunctive amplification) 4. $P \rightarrow Q$ ($\neg P \vee Q \Leftrightarrow P \rightarrow Q$)	½ ½ 1 1	3

II 3)	Relation $R = \{(1,1), (2,2), (3,3), (4,4), (1,2)\}$ R is reflexive and transitive But not Symmetric Since $(1,2) \in R$ but $(2,1) \notin R$	① ① ①	3
II 4)	ϕ, $\{1\}$, $\{2\}$, $\{3\}$, $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, $\{1,2,3\}$	3	3
II 5)	$f(x) = \sin x$ $\quad g(x) = 5x^2$. $gof(x) = g(f(x))$ $= g(\sin x)$ $= 5(\sin x)^2$	1 1 1	3
II 6)	$R = \{(2,1), (3,1), (4,1), (3,2), (4,2), (4,3)\}$ Graph of R.  Matrix Representation $M_R = \begin{cases} 1 & \text{if } (x,y) \in R \\ 0 & \text{if } (x,y) \notin R \end{cases}$ Matrix. $= \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$	1 1 1	3

II 7.	In the word MATHEMATICS, we treat the vowels AEAI as one letter. MTHMTICS (AEAI) → 8 letters. ↓ Here M comes twice, T comes twice In AEAI → A come twice. The number of different ways $= \frac{8!}{2!\,2!} \times \frac{4!}{2!} = 120960$	1 ½ ½ 1	3	
II 8.	$(k+1)^{th}$ term of $(x+a)^n = {}^nC_r\, x^{n-r} a^r$. 10th term of $\left(x^2 + \frac{1}{x^2}\right)^{20}$ $= {}^{20}C_9 \left(x^2\right)^{20-9} \left(\frac{1}{x^2}\right)^9$ $= {}^{20}C_9 \left(x^2\right)^{11} \times \frac{1}{x^{18}}$ $= {}^{20}C_9\, x^4$	1 1 1	3	
II 9.	① for any $a, b \in Z$, $a+b \in Z$ $\Rightarrow$ + is closed. ② for any $a, b, c \in Z$, $(a+b)+c = a+(b+c)$ $\Rightarrow$ + is associative ③ for any $a \in Z$, $a+0 = 0+a = a$ $\Rightarrow$ Z has identity 0 $\therefore (Z, +)$ is monoid.	1 1 1	3.	

II 10. Let $a, b \in H \cap K$

Then $a, b \in H$ and $a, b \in K$ — 1

$\Rightarrow H \cap K$ is closed

Then $e \in H$ and $e \in K$ (H and K are Subgroups)

$\Rightarrow$ (identity) e exist in $H \cap K$ — 1 — 3

Let $a \in H \cap K$, Then $a \in H$ and $a \in K$

$\Rightarrow a^{-1} \in H$ and $a^{-1} \in K$

$\Rightarrow a^{-1} \in H \cap K$ ie inverse exist in $H \cap K$ — 1

$\therefore H \cap K$ is a Subgroup.

II PART C — 6x7

III 1.

P	q	r	s	$q \to s$	(a) $p \to (q \to s)$	(b) $\neg r \vee p$	$q \wedge p \wedge q$
T	T	T	T	T	T	T	T
T	T	T	F	F	F	T	F
T	T	F	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	T	F
T	F	T	F	T	T	T	F
T	F	F	T	T	T	T	F
T	F	F	F	T	T	T	F
F	T	T	T	T	T	F	F
F	T	T	F	F	T	F	F
F	T	F	T	T	T	T	T
F	T	F	F	F	T	T	T
F	F	T	T	T	T	F	F
F	F	T	F	T	T	F	F
F	F	F	T	T	T	T	F
F	F	F	F	T	T	T	F

III — 7 — 7

III 2. $p: -1 < 3$ TRUE

$q: 3+7=10$ TRUE

$r: \sin\left(\frac{3\pi}{2}\right) = -1$ TRUE

The given statement is $(p \wedge q) \rightarrow r$.

$p \wedge q$ is TRUE

Converse is $r \rightarrow (p \wedge q)$

$1 \rightarrow 1$ is 1 TRUE

Inverse is $\neg(p \wedge q) \rightarrow \neg r$

$F \rightarrow F$

$0 \rightarrow 0$ is TRUE

Contra positive is $\neg r \rightarrow \neg(p \wedge q)$

$0 \rightarrow 0$ is TRUE

7

III 3.

Steps	Reasons.
1) $\neg s \wedge \neg u$	Premise.
2) $\neg u$	① Simplification
3) $\neg u \rightarrow \neg t$	Premise.
4) $\neg t$	② ③ Modus ponens.
5) $\neg s$	① Simplication
6) $\neg s \wedge \neg t$	④ ⑤ Conjunction
7) $r \rightarrow (s \vee t)$	Premise.
8) $\neg(s \vee t) \rightarrow \neg r$	⑦ $r \rightarrow (s \vee t) \Leftrightarrow \neg(s \vee t) \rightarrow \neg r$
9) $(\neg s \wedge \neg t) \rightarrow \neg r$	⑧ De Morgans law.

	10) $\neg r$	⑧⑨ Modus Ponens	½		
	11) $(\neg p \vee q) \rightarrow r$	premise.	1		
	12) $\neg r \rightarrow \neg(\neg p \vee q)$	⑪ Contrapositive		7	
	13) $\neg r \rightarrow (p \wedge \neg q)$	⑫ De Morgan's law	1		
	14) $p \wedge \neg q$	⑩⑬ Modus Ponens.			
	15) $\therefore p$	⑭ Simplification			

III 4)	Rule of Inference	Related logical Implication	Name of Rule.		
	1) p, $p \rightarrow q$ / $\therefore q$	$[p \wedge (p \rightarrow q)] \rightarrow p$	Rule of detachment (Modus Ponens)	1	
	2) $p \rightarrow q$, $q \rightarrow r$ / $\therefore p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Law of the Syllogism.	1	
	3) $p \rightarrow q$, $\neg q$ / $\therefore \neg p$	$[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$	Modus Tollens	1	7
	4) p, q / $\therefore p \wedge q$	–	Rule of Conjunction	1	
	5) $p \vee q$, $\neg p$ / $\therefore q$	$[(p \vee q) \wedge \neg p] \rightarrow q$	Rule of Disjunctive Syllogism.	1	

	6) $\neg P \rightarrow F_0$ $\therefore P$.	$(\neg P \rightarrow F_0) \rightarrow P$	Rule of Contradiction	1	
	7) $P \wedge Q$ $\therefore P$	$(P \wedge Q) \rightarrow P$	Rule of Conjunctive Simplification	1	
III 5. a)	let P → No of Students who study Physics E → " " English. B → " " Biology. No of Students in the group = $n(P \cup E \cup B)$ Given $n(P) = 49$ $n(E) = 37$ $n(B) = 21$ $n(P \cap E) = 9$ $n(E \cap B) = 5$ $n(P \cap B) = 4$ $n(P \cap E \cap B) = 3$ $n(P \cup E \cup B) = n(P) + n(E) + n(B)$ $- n(P \cap E) - n(E \cap B) - n(P \cap B)$ $+ n(P \cap E \cap B)$ $= 49 + 37 + 21 - 9 - 5 - 4 + 3$. $= 92$.			2. 1 1	4.

III 5 b) $y = f(x) = 6x + 5$ — ½

$y - 5 = 6x$

$x = \frac{y-5}{6}$ — ½

Inverse of f is $f^{-1}(x) = \frac{x-5}{6}$ — 1

given $f(x) = 6x+5$ $\quad f^{-1}(x) = \frac{x-5}{6}$

$\therefore f \circ f^{-1}(x) = f(f^{-1}(x)) = f\left(\frac{x-5}{6}\right)$ — ½

$= 6\left(\frac{x-5}{6}\right) + 5$ — ½

$= x$. — ½

Similarly $f^{-1} \circ f(x) = x$.

III 6) i) Inverse Relation : Let R be a relation from Set A to Set B, then the inverse of R is the relation from B to A. and is given by $R^{-1} = \{(y, x) / (x, y) \in R\}$ — 1

eg: If $R = \{(1,2), (3,2), (5,1)\}$

$R^{-1} = \{(2,1), (2,3), (1,5)\}$

2) Universal Relation: A relation R on a Set A is called universal relation if $R = A \times A$ — 1

eg: If $A = \{1, 2, 3\}$

Then $R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

3) Empty relation: A relation R on a set A is called empty relation if $R = \phi$

eg: If $A = \{3, 4\}$ and $R = \{(x, y) / x + y > 9\}$

$\therefore R = \phi$

4) Identity relation: A relation R on a set A is called identity relation if

$R = \{(x, x) / x \in A\}$.

eg: let $A = \{1, 2\}$

Relation $R = \{(1, 1), (2, 2)\}$

5) Reflexive relation: A relation R on a Set A is called reflexive. if $(x, x) \in R$ for all $x \in A$

eg: $A = \{1, 2\}$

Relation $R = \{(1, 1), (1, 2), (2, 2)\}$.

6) Symmetric relation:

If $(x, y) \in R$ then $(y, x) \in R$ for all $x, y \in A$.

eg: $A = \{1, 2, 3\}$

$\{(1, 1), (2, 1), (1, 2), (2, 3), (3, 2), (1, 3), (3, 1), (2, 2), (3, 3)\}$.

7) Transitive relation: If $(x, y) \in R$

7

and $(y, z) \in R$ then $(x, z) \in R$ for all $x, y, z \in A$. 1

eg: $A = \{1, 2\}$

$R = \{(1,1), (1,2), (2,1), (2,2)\}$

III 7) (i) No of ways to select all five ball are red $= {}^8C_5 = \frac{8!}{(8-5)!5!} = \underline{56}$ 1

(ii) No of ways to select 2 red balls out of 8 red balls $= {}^8C_2$. 1

$= 28$

No of ways to select 3 black balls out of 7 black balls $= {}^7C_3$ 1

$= 35$

∴ No of ways to select 2 red and 3 black balls $= 28 \times 35$ ½.

$= \underline{980}$

(iii) No of ways to select atleast 3 red balls = 3

$${}^8C_3 \times {}^7C_2 + {}^8C_4 \times {}^7C_1 + {}^8C_5 \times {}^7C_0$$

$= 1176 + 490 + 56$

$= \underline{1722}$. ½

III 8) The generating function is

$a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \cdots + a_n t^n + \cdots$

$$G(t) = \sum_{r=0}^{\infty} a_r t^r \quad (1)$$

given $a_{r+2} - 3a_{r+1} + 2a_r = 0$ ——— (1)

Multiply eqn (1) by t^r and Summing from $r = 0$ to ∞, we get

$$\sum_{r=0}^{\infty} a_{r+2} t^r - 3\sum_{r=0}^{\infty} a_{r+1} t^r + 2\sum_{r=0}^{\infty} a_r t^r = 0 \quad (1)$$

$$(a_2 + a_3 t + a_4 t^2 + \cdots) - 3(a_1 + a_2 t + a_3 t^2 + \cdots) + 2(a_0 + a_1 t + a_2 t^2 + \cdots) = 0 \quad \text{——— (2)}$$

Since $G(t) = a_0 + a_1 t + a_2 t^2 + \cdots$ ½

$$(2) \Rightarrow \frac{G(t) - a_0 - a_1 t}{t^2} - 3\left(\frac{G(t) - a_0}{t}\right) + 2G(t) = 0 \quad \text{——— (3)}$$ ½

Put $a_0 = 2$ and $a_1 = 3$ in (3) we get

$$\Rightarrow G(t) = \frac{2 - 3t}{1 - 3t + 2t^2}.$$

$$G(t) = \frac{2 - 3t}{(1-t)(1-2t)} \quad 1\frac{1}{2}$$

By partial fraction: $\frac{2-3t}{(1-t)(1-2t)} = \frac{A}{1-t} + \frac{B}{1-2t}$ ½

$$\frac{2-3t}{(1-t)(1-2t)} = \frac{A(1-2t) + B(1-t)}{(1-t)(1-2t)}$$ ½

$$2-3t = A(1-2t) + B(1-t)$$

Put $t=1$, $-1 = -A \Rightarrow A = 1$ ½

Put $t=\frac{1}{2}$ $\frac{1}{2} = \frac{1}{2}B \Rightarrow B = 1$ ½

$\therefore G(t) = \frac{1}{1-t} + \frac{1}{1-2t}$

Hence $a_r = 1 + 2^r$ 1 — F

III 9) Recurrence relation $a_{n+2} = a_{n+1} + a_n$.

$\Rightarrow a_{n+2} - a_{n+1} - a_n = 0$ 1

Characteristic equation is $R^2 - r - 1 = 0$ 1

$\therefore R = \frac{1 \pm \sqrt{5}}{2}$ 1

$\therefore$ General solution is

$$a_n = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n.$$ 1 — F

$a_0 = 0 \Rightarrow c_1\left(\frac{1+\sqrt{5}}{2}\right)^0 + c_2\left(\frac{1-\sqrt{5}}{2}\right)^0 = 0$

$c_1 + c_2 = 0 \Rightarrow c_1 = -c_2$ 1

$a_1 = 1 \Rightarrow c_1\left(\frac{1+\sqrt{5}}{2}\right) + c_2\left(\frac{1-\sqrt{5}}{2}\right) = 1$

$\frac{c_1 2\sqrt{5}}{2} = 1 \Rightarrow c_1 = \frac{1}{\sqrt{5}}$ 1

$c_2 = -\frac{1}{\sqrt{5}}$

$\therefore$ General solution is $a_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$ 1

III 10) **Binomial Theorem:-**

If x and y are variables and n is a positive integer then.

$$(x+y)^n = nC_0\, x^0 y^n + nC_1 x^1 y^{n-1} + nC_2 x^2 y^{n-2} + \cdots + nC_n x^n y^0 \quad \text{2.}$$

$$= \sum_{k=0}^{n} nC_k\, x^k y^{n-k}.$$

given $(x+2y)^7$

Since $n = 7$ is odd, the middle terms are. $\frac{n+1}{2} = \frac{7+1}{2} = 4^{th}$ and $\frac{n+3}{2} = 5^{th}$ terms. 1

7.

4th term $= (3+1)^{th}$ term $= 7C_3\, x^{7-3} (2y)^3$

$= 35\, x^4 \cdot 8 \cdot y^3$ 2

$= 280\, x^4 y^3.$

5th term $= (4+1)^{th}$ term $= 7C_4 (x)^{7-4} (2y)^4$.

$= 35\, x^3 \times 16\, y^4$ 2.

$= 560\, x^3 y^4.$

III 11) Given $a * b = a + b + 2$

1) Closure property :- Let $a, b \in Z$.

$a * b = a + b + 2$ is an integer. 1

$\therefore a * b \in Z \Rightarrow *$ is a closure.

2) Associative property : let $a, b, c \in Z$

$a*(b*c) = a*(b+c+2)$

$= a+b+c+2+2 = a+b+c+4$ — 2

$(a*b)*c = (a+b+2)*c$

$= a+b+2+c+2 = a+b+c+4$

$\therefore a*(b*c) = (a*b)*c$

$\Rightarrow *$ is associative

3) Existence of identity.

$a*e = a \Rightarrow a+e+2 = a$

$\Rightarrow e = a-a-2 = -2.$ — 1½

$e = -2 \in Z$.

$\therefore$ there exist identity $e = -2 \in Z$. such that $a*-2 = a$ for $a \in Z$

4) Existence of inverse

Let a^{-1} be the inverse of a in Z.

$a*a^{-1} = -2 \Rightarrow a+a^{-1}+2 = -2.$

$a^{-1} = -2-2-a$ — 1½

$a^{-1} = -4-a \in Z$

$\therefore$ There exist inverse $a^{-1} \in Z$ such that

$a*a^{-1} = e = -2.$

From 1, 2, 3, and 4 we get $(Z, *)$ is a group. — 1

III 12 a) Let (X, o) and $(Y, *)$ be two algebraic System where o, and $*$ are both binary

operations. A function $f: X \to Y$ is known as homomorphism from $(X, \circ)$ to $(Y, *)$ if for any $a, b \in X$ we have.

$f(a \circ b) = f(a) * f(b)$ — 2

Given $g(a) = a \pmod 4$ — 4

Let $a, b \in \mathbb{N}$

$g(a+b) = a+b \pmod 4$ — ½

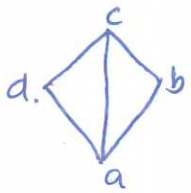
$= a \pmod 4 +_4 b \pmod 4$ — ½

$= g(a) +_4 g(b)$

$\Rightarrow g$ is a homomorphism. — 1

III 12b) 1) Bounded lattice: If it has the greatest element 1 and least element 0. — 1

2) eg:



2) Complete lattice: A lattice is complete if each of its non-empty subset hav a least upper bound and greatest lower bound. — 1

3) Complemented lattice: A lattice L is called a Complimented lattice if L is bounded and every element in L has a Complement. — 1